

Program: Diploma in Civil and Environmental Engineering	
Course Code: 4359	Course Title: Irrigation Engineering Lab
Semester: 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Impart basic knowledge of irrigation structures to the students
- Enable the students to prepare irrigation drawings

Course Prerequisites:

Topic	Course code	Course name	Semester
Engineering Graphics		Engineering Graphics	1
Elementary Mathematics		Engineering mathematics	1&2
Building drawing		Civil Engineering Drafting Lab	3

Course Outcomes:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Sketch irrigation structures and culvert	14	Applying
CO2	Measurement of physical characteristics of soil.	11	Applying
CO3	Calculation on irrigation water needs	9	Applying
CO4	Familiarize with local irrigation structures and practices	7	Applying
	Lab Tests	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3			3			
CO3	3			3			
CO4							3

3-Stronglymapped, 2-Moderatelymapped,1-Weaklymapped

CourseOutline

On completion of the course student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Sketch irrigation structures and culvert		
M1.01	Sketch the cross section of earth dam and masonry showing component parts	1	Applying
M1.02	Sketch the views of a canal drop showing component parts	1	Applying
M1.03	Sketch the views of surplus escape, showing all components	2	Applying
M1.04	Sketch the views of a weir showing component parts	2	Applying
M1.05	Sketch the views of a Tank sluice with tower head	2	Applying
M1.06	Sketch the given canal section	2	Applying
M1.07	Draw a labeled sketch of the given culvert (without splayed wing wall)	2	Applying
M1.08	Sketch the views of aqueduct showing component parts	2	Applying

CO2	Measurement of physical characteristics of soil.		
M2.01	Soil moisture measuring devices : To study about the following soil moisture measurement devices-Soil sampling augers- Neutron probe- Time domain reflectometer (TDR)- Electrical resistance block/ Gypsum blocks- Tensiometer- Pressure membrane or pressure plate apparatus	2	Understanding
M2.02	Measurement of soil moisture by different soil moisture measuring instruments	3	Applying
M2.03	Measurement of infiltration characteristics by using double ring infiltrometer	3	Applying
M2.04	Determination of bulk density, field capacity and wilting point.	3	Applying
	Lab Test-I	2	
CO3	Calculation on irrigation water needs		
M3.01	Measurement of irrigation water using weir, flume and orifice.	2	Applying
M3.02	Methods to measure the evapotranspiration rates of crops - Evaporation pan and Using equations that predict the evapotranspiration rate based on climatic parameters.	2	Understanding
M3.03	Measurement of evapotranspiration	3	Applying
M3.04	Sketch the layout of sprinkler irrigation and drip irrigation	2	Applying
CO4	Familiarize with local irrigation structures and practices		
M4.01	Open ended experiment	7	Applying
	Lab Test II	2	

Text/Reference:

T/R	BookTitle/Author
T1	C. Punmia : Irrigation Engineering ; Laxmi Publishing Co:
T2	Modi &Sethi : Irrigation Engineering ; Standard Publishing House
R1	S.K. Garg : Irrigation Engineering ; Khanna Publishers
R2	B S Birdie : Irrigation Engineering & Water Power Engg ; Standard Publishing Hous
R3	N.N. Basak : Irrigation Engineering ; McGraw Hill Publishing

Online Resources:

Sl. No.	Website Link
1	https://nptel.ac.in/
2	http://egyankosh.ac.in/
3	https://swayam.gov.in/